

Breakout

Project Plan
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1 Project description

Breakout is a single-player 2D arcade game. The player moves a paddle horizontally to keep a ball in play and destroy a grid of bricks. A session ends when the player clears every brick or loses all three lives. The game includes a start screen, gameplay screen, win screen and game over screen.

1.1 Problem and intended users

The project challenge is to turn a familiar game concept into a responsive and understandable application. The player needs predictable ball behaviour, controls that are easy to learn, and visible score and life information. For me as the developer, the challenge is coordinating real-time movement, collisions and game states in a maintainable Java application.

The intended audience is casual players who want a short arcade session, including players with little gaming experience. This audience description comes from the game concept and still needs to be checked through playtesting.

2 Scope and constraints

The first version targets Windows and contains one fixed play area with one brick layout. It uses keyboard controls, a visible score, three lives and restart functionality. The plan is to use Java and LibGDX. Java is already my strongest language, so the technology comparison should consider how much time each option leaves for learning game development.

Multiplayer, a level editor, saved progression, selectable difficulty, additional levels and configurable settings are outside this version. Optional audio must not delay the playable core. Development time is limited by the course schedule, so the Advice document proposes a relative plan that can be aligned with the submission date.

3 Risks and validation priorities

Risk	Effect	Response
Missed or repeated collisions	The ball passes through a brick or becomes stuck.	Test side and corner contacts and supported speeds. Correct contact position and review update steps if tunnelling occurs.
Incorrect state reset	New games retain old score, lives or objects.	Centralise session reset behaviour and repeat the REQ-11 and Q02 checks.
Scope growth	Optional work reduces time for core testing.	Complete Must haves before angle tuning or audio. Keep future levels outside scope.
Unverified generated code	Code may be incorrect or difficult to explain.	If AI is used, review small changes, check documentation and test each increment.

4 Project Planning

The table below records how the project actually progressed across the first two weeks, structured around the LO2 activities (Analyse, Advise, Design, Realise, Validate). Unlike the original proposed planning, this reflects real progress rather than an estimate.

LO2 activity	Timing	Status and evidence
Analyse	Week 1, first part	Completed. Problem, scope and requirements defined; see Breakout Analysis.
Advise	Week 1, first part – Week 2, first part	Completed. Technology comparison and proposed solution confirmed; see Breakout Advice.
Design	Week 1, second part – Week 2, first part	Completed. Started with a first prototype of the wireframes and UML class diagram (Week 1, second part), then finalised the wireframes and produced an updated UML diagram covering all classes and methods (Week 2, first part); see Breakout Design.
Realise	Week 2, second part – Week 3, first part	Started. Implementation (coding) began in the second part of Week 2.
Validate	Week 4	Planned once enough of Realise is complete to test against the acceptance checks in Breakout Analysis.

4.1 Milestones

By the end of the first part of Week 1, Analysis and Advice had both started. By the end of the second part of Week 1, Design had started, Analysis was finished, and a first prototype of the wireframes and class diagram existed. By the end of the first part of Week 2, Design and Advice were both finished, with a final wireframe set and a UML diagram covering all classes and methods. The second part of Week 2 marks the start of Realise (coding).

4.2 Design Evolution

The figures below show the prototype wireframes and UML class diagram produced in the second part of Week 1. See Breakout Design for the final wireframes and the final UML class diagram, produced in the first part of Week 2 with all classes and methods included.



